

实验五：stateflow 中流程图的建模与仿真

实验目的：

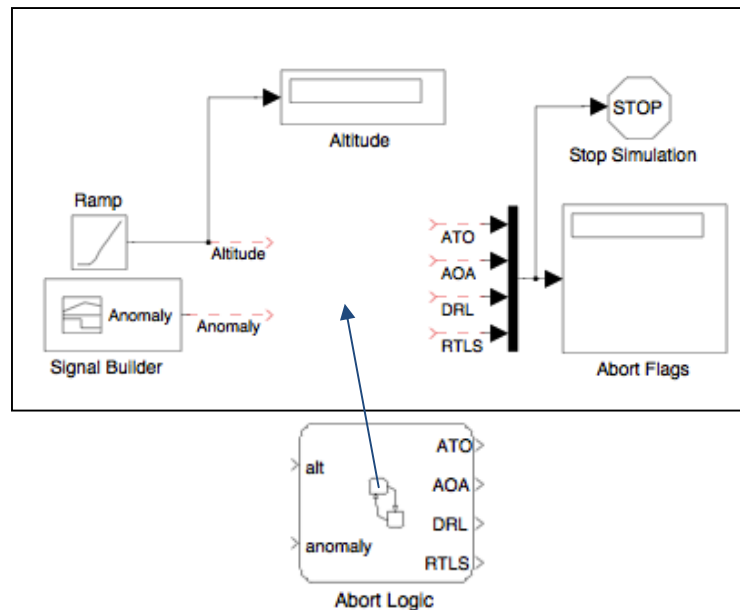
熟悉如何使用 simulink 中 stateflow library，掌握如何为复杂的逻辑结构建立流程图，并对此进行仿真。掌握在 stateflow 中对于状态机建模的初步。熟悉 debug 的使用。

实验内容：

1. In this exercise, you will implement logic for a launch abort system. A launch abort system is designed to abort the launch of a space shuttle if anomalies are detected. The altitude of the space shuttle at the time of the abort determines how the vehicle will return safely to the ground. The possible situations are
 - (1) If altitude is greater than or equal to 50000m, abort into orbit(ATO), meaning the shuttle is released into orbit.
 - (2) If altitude is greater than or equal to 10000m, but less than 50000, abort once around(AOA).
 - (3) If altitude is greater than or equal to 1000m, but less than 10000, abort to a downrange landing(DRL).
 - (4) If altitude is less than 1000, abort and return to launch site (RTLS)

Open the model abort_logic_start and create a Stateflow chart that will model the logic described above. Pseudocode for this logic can be written as follows:

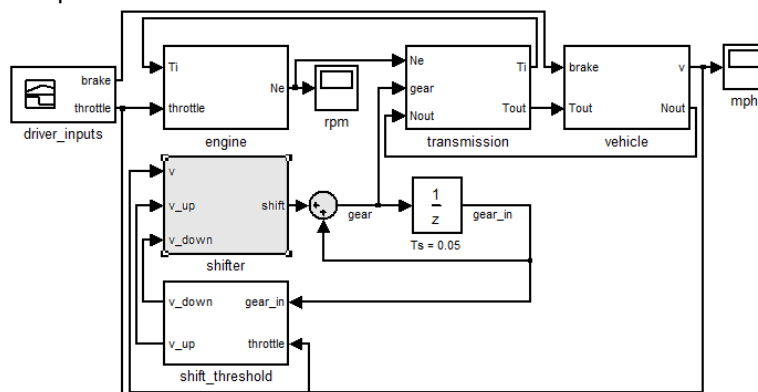
```
ATO=0
AOA=0
DRL=0
RTLS=0
if Anomaly==1
    if Altitude>=50000
        ATO=1
    else if Altitude>=10000
        AOA=1
    else if Altitude>=1000
        DRL=1
    else
        RTLS=1
end
```



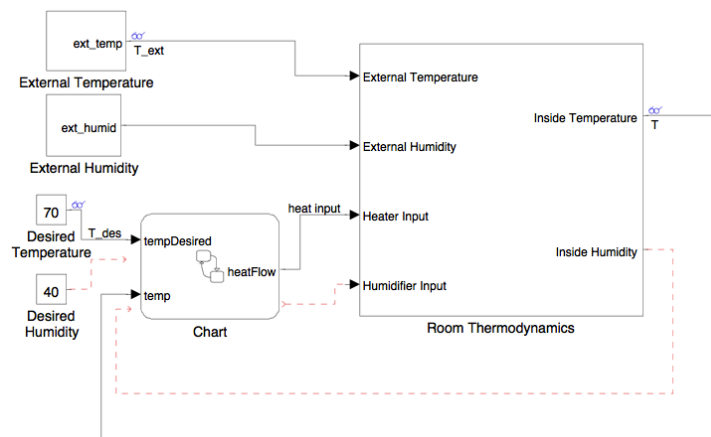
2. Open the simplified vehicle model *car* and study the mechanism that handles the shifting of the automatic transmission. Notice that the shifting control is running at a rate of 0.05. The control logic relies on speed thresholds returned from lookup tables to govern the gear shift. The logic shifts to a higher gear when the vehicle speed exceeds an upshift shreshold. Likewise, it shifts to a lower gear when the vehicle speed drop below a downshift shreshold. The new gear value is then derived by accumulating the shifting command to the existing value.

Redesign the model by creating a state machine to implement the transmission gears such that the *shifter subsystem* and the *Sum block* that maintains the gear value can be eliminated. The state machine should

- . Maintain four gears as exclusive states.
- . Receive the vehicle speed, upshift threshold, and downshift threshold as inputs.
- . Return the gear value as an output.
- . Sample at 0.05.



3. In the following model, the chart is currently designed to enable the cooler and heater based on the room temperature and the desired temperature setting. Modify the state machine in the file `climateCtr1` such that the chart controls the humidifier(加湿器) as well. The humidifier output is `dh`, the change in humidity. Model the humidifier control using secondary states in the chart. The new implementation should have the following behavior:
- (1). The humidifier should be enabled when the room humidity drops below the desired setting.
 - (2). The humidifier can be enabled only if either the cooler or heater is enabled.
 - (3). the humidifier is disabled when the cooler or heater is initially enabled.
 - (4). when the humidifier is enabled, the humidifier input to the Room Thermodynamics subsystem should be set to 1.5, otherwise it should be set to 0.



4. In this exercise, we will use the multiprocessing or parallel, capabilities of the Stateflow software to modify the climate control algorithm. The additions we will make will allow the control strategy to toggle between aggressive and non aggressive.

(1) Open the starting model –temParallelStart.

(2) Modify the chart to contain two parallel start, A and B

State A-This state should contain all of the current contents of the state chart.

State B-this state should contain the following logic, implemented using states:

```
if (aggressive==1)
    dt=0.1; //aggressive state
else
    dt=2;    //nonaggressive state
```

(3) Set the execution order of the two parallel states such that state B executes before state A when the chart is updated.

